LYKA A. DINGLASAN

Bachelor of Science in Electronics Engineering

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A highly motivated BS Electronics Engineering graduate from LPU-Cavite, equipped with a strong foundation in analog and digital circuit design, embedded systems, and programming. Experienced in developing innovative electronic projects, from logic circuit PCBs to automation and monitoring systems, demonstrating a keen ability to design, analyze, and optimize electronic solutions. Adept at troubleshooting, problem-solving, and collaborating.

EDUCATION

Bachelor of Science in Electronics Engineering
2022 - 2026
Lyceum of the Philippines University – Cavite
Dean's Lister (2022 – 2023)
Dean's Lister (2023 – 2024)

Senior High School – STEM Strand
2020 – 2022
Christian Grace School of Cavite
Valedictorian

Internship — **July 2025 - August 2025**
Northstar Technologies Industrial Corporation
Completed a 240-hour engineering practicum in the Product Design and Development and Product Board Manufacturing departments, performing BGA rework, thermal profiling, and impedance diagnostics on semiconductor test boards. Assembled and tested Javelin J800 MX1, EX, and MX2 semiconductor test systems.

CERTIFICATIONS

- TOEIC Listening and Reading Test – 975/990 (2026)
- Safety Officer 2 (SO2) – Basic Occupational Safety and Health Training Certification (2025)
- Google Cybersecurity Certification (2026)
- Computer System Servicing NC II (2026)

AWARDS

Laurus Nobilis Quality Awards (2026)
Lyceum of the Philippines University - Cavite

DOST-FPRDI LAWIG DISENYO Furniture Design Contest (2024) - Finalist

LPU - Cavite Physilympics (2024) - 1st Place

Commquest 2025: Communications Quiz Challenge (2025) - 3rd Runner Up

IECEP Cavite Chapter Digital Poster Making Competition (2024) - 1st Place

SKILLS

- **Circuit Design:** Analog Circuit Design, Digital Logic, Embedded Systems
- **Testing and Instrumentation:** Electronic Test and Measurement, Laboratory Instrumentation
- **Programming Languages:** C, C++, Python, JavaScript, MATLAB
- **PCB and Design:** PCB Layout and Design, Schematic Interpretation
- **PCB and Circuit Design Tools:** KiCad, LTspice, NI Multisim, Tinkercad
- **Soft Skills:** Analytical troubleshooting, creative problem-solving, debugging, clear technical writing, teamwork and collaboration
- **Simulation and Analysis Tools:** ATmega (Arduino), MATLAB, GNU Octave, Scilab, LabVIEW, PicoSoft, Cisco Packet Tracer
- **3D CAD and Modeling:** Solidworks, AutoCAD, SketchUp

LEADERSHIP EXPERIENCE

Creative Director — 2024-2025
IECEP Cavite Electronics Engineering Student Chapter

Director of Publications — 2024-2025
Jr. Institute of Electronics Engineers LPU – Cavite Student Chapter

LEADERSHIP EXPERIENCE

Creatives Committee 2022-2023
Member of the Creatives Committee for LPU Cavite - College of Engineering and Architecture Student Government

Editor-in-Chief 2022-2023
CGS Publication Society, Christian Grace School of Cavite

Vice-President 2018 – 2019
Math-Sci Club, Christian Grace School of Cavite

Secretary 2017 – 2018
CommArts Club, Christian Grace School of Cavite

PROJECT HISTORY

ALERSENSE: Wearable Physiological Response Sensors and YOLOv12-Based Attention Monitoring with Haptic Feedback

Technologies Used: Raspberry Pi 5, YOLOv12, Firebase, Python, Roboflow

- Developed a wearable attention-monitoring system tracking Heart Rate, Galvanic Skin Response, and Skin Temperature, paired with a YOLOv12-powered camera module that recognizes inattentive classroom behaviors in real time. Delivers instant haptic feedback and updates a live teacher dashboard the moment both systems confirm inattention, bridging biometric data with classroom intervention.

Milk Tea Shop POS and Inventory Management System

Technologies Used: React, JavaScript, Vite, Redis

- Built a full-stack POS and inventory management system (React/Vite, Upstash Redis) with sales transactions, inventory tracking, sales analytics, and low-stock alerts.

Cabling System for Call Center Building

Technologies Used: AutoCAD, Cisco Packet Tracer, Cabling Design

- Designed a cabling infrastructure and network device layout for a multi-story call center building, including backbone/horizontal cabling, telecom rooms, and workstation connectivity.

Smart Home Automation System

Technologies Used: Cisco Packet Tracer, Internet of Things (IoT), Network Simulation

- Designed and simulated an IoT smart home integrating intelligent appliances, environmental sensors, renewable energy, and cloud connectivity for real-time monitoring and remote control of household devices.

LabVIEW-Based Smart Home Energy Monitoring System

Technologies Used: LabVIEW, Smart Home, Energy Monitoring

- Developed a LabVIEW application with an interactive dashboard to monitor and control household electrical loads, visualizing voltage, current, power, energy consumption, and appliance runtime in real time.

LabVIEW-Based Function Generator and Oscilloscope

Technologies Used: LabVIEW, Signal Processing, Virtual Instrumentation, Data Acquisition

- Designed a LabVIEW virtual instrument generating sine, square, triangle, and sawtooth waveforms with real-time oscilloscope visualization, adjustable amplitude/frequency/phase/duty cycle, triggering, and waveform statistics.

PAM, PWM, and PPM Signal Generator

Technologies Used: KiCad, Circuit Design, PCB Design

- Designed and fabricated a KiCad-based PCB generating PAM, PWM, and PPM signals using NE555 timer circuits with a DIP switch interface for selecting modulation modes.

REFERENCE

Dr. Leah Q. Santos
BS-ECE Program Chair, LPU Cavite - College of Engineering and Architecture
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Dr. Den Whilrex Garcia
BS-ECE Faculty, LPU Cavite - College of Engineering and Architecture
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